

MARSEILLE PROVENCE AP, France

WMO: 076500

Lat: 43.438N

Lon: 5.216E

Elev: 23

StdP: 101.05

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-2.2	-0.8	-8.2	1.9	1.0	-6.1	2.3	2.1	17.9	7.9	15.8	8.0	2.8	140	0.530

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.5	33.1	21.0	31.8	20.7	30.6	20.4	23.2	29.3	22.5	28.7	21.8	28.0	5.9	260

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
21.3	16.0	26.4	20.4	15.1	25.8	19.6	14.4	25.4	69.0	29.4	66.1	28.7	63.5	27.9	26.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
15.8	13.8	12.0	-4.3	35.6	1.8	1.7	-5.6	36.8	-6.6	37.7	-7.6	38.7	-9.0	39.9
			-5.2	24.5	1.8	0.7	-6.5	25.0	-7.6	25.4	-8.6	25.8	-10.0	26.3
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	15.8	7.5	8.1	11.3	14.3	18.3	22.6	25.2	24.7	20.7	17.0	11.5	8.3	(6)
(7)		DBStd	6.92	3.30	3.28	2.77	2.58	2.67	2.91	2.28	2.23	2.63	3.14	3.69	3.53	(7)
(8)		HDD10.0	280	89	67	18	2	0	0	0	0	0	1	26	76	(8)
(9)		HDD18.3	1588	334	287	218	123	33	3	1	0	9	62	206	313	(9)
(10)		CDD10.0	2410	13	13	59	131	256	377	470	457	321	219	71	22	(10)
(11)		CDD18.3	675	0	0	0	2	31	130	212	199	79	21	1	0	(11)
(12)		CDH23.3	5632	0	0	1	15	200	1086	2028	1773	469	61	0	0	(12)
(13)		CDH26.7	1938	0	0	0	1	32	353	798	659	92	3	0	0	(13)
(14)	Wind	WSAvg	4.8	4.5	4.9	5.0	5.2	5.0	4.9	5.2	4.7	4.6	4.4	4.8	4.4	(14)
(15)	Precipitation	PrecAvg	501	46	32	26	49	37	25	11	22	77	72	66	40	(15)
(16)		PrecMax	734	136	117	72	136	88	110	66	73	210	142	163	150	(16)
(17)		PrecMin	236	0	0	0	0	0	0	0	0	0	23	5	0	(17)
(18)		PrecStd	138	39	30	21	33	27	28	14	21	56	32	43	41	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	16.9	17.9	21.8	25.1	29.9	34.1	34.7	35.2	30.8	26.8	21.1	17.6	(19)
(20)		0.4%	MCWB	11.9	11.6	13.2	15.9	18.7	21.1	21.0	21.5	19.9	19.0	16.4	13.7	(20)
(21)		2%	DB	15.1	16.1	19.6	22.9	27.2	32.1	33.1	33.0	29.1	25.0	19.5	16.1	(21)
(22)		2%	MCWB	11.4	10.8	12.4	15.0	17.2	20.6	20.9	21.1	19.7	18.5	15.5	12.9	(22)
(23)		5%	DB	14.0	14.7	18.0	21.2	25.5	30.2	31.9	31.5	27.5	23.6	18.4	15.0	(23)
(24)		5%	MCWB	10.9	10.3	11.9	13.8	16.6	20.0	20.6	20.9	19.1	17.9	14.8	12.1	(24)
(25)		10%	DB	12.9	13.4	16.6	19.7	24.0	28.7	30.8	30.2	26.1	22.2	17.2	13.8	(25)
(26)		10%	MCWB	9.9	9.7	11.3	13.1	16.1	19.3	20.4	20.6	18.5	17.2	14.0	11.1	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	13.1	12.6	14.4	17.0	20.1	23.3	24.2	24.2	22.4	20.5	17.5	14.6	(27)
(28)		0.4%	MCDWB	15.4	15.9	19.4	23.3	26.9	31.0	29.6	29.8	27.1	24.5	19.9	16.6	(28)
(29)		2%	WB	12.1	11.8	13.4	15.6	18.6	22.1	23.2	23.3	21.4	19.6	16.2	13.4	(29)
(30)		2%	MCDWB	14.5	14.9	17.8	21.4	24.8	29.1	29.4	29.1	26.2	23.5	18.8	15.5	(30)
(31)		5%	WB	11.1	11.0	12.6	14.7	17.7	21.1	22.4	22.6	20.6	18.7	15.3	12.4	(31)
(32)		5%	MCDWB	13.6	14.0	16.7	19.9	23.4	28.0	28.9	28.5	25.3	22.6	18.0	14.6	(32)
(33)		10%	WB	10.2	10.1	11.9	13.9	16.9	20.2	21.6	21.9	19.7	17.8	14.3	11.4	(33)
(34)		10%	MCDWB	12.6	12.9	15.8	18.4	22.3	27.0	28.3	27.8	24.5	21.4	17.0	13.6	(34)
(35)	Mean Daily Temperature Range		MDBR	7.9	8.8	9.6	9.5	9.8	10.3	10.5	9.7	8.7	8.0	7.7	(35)	
(36)		5% DB	MCDBR	8.2	10.3	11.5	11.6	11.9	11.9	11.8	12.1	11.4	9.6	8.2	7.8	(36)
(37)			MCWBR	5.6	6.5	6.4	5.9	5.4	4.8	4.7	5.0	5.2	5.0	5.2	5.3	(37)
(38)			MCDBR	7.3	9.0	10.0	10.5	10.5	10.9	10.5	10.5	9.5	8.3	7.3	6.8	(38)
(39)		5% WB	MCWBR	5.5	6.2	6.0	5.9	5.3	4.9	4.9	5.3	5.1	4.8	5.1	5.1	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.321	0.338	0.372	0.396	0.406	0.405	0.399	0.397	0.386	0.380	0.345	0.317	(40)	
(41)		taud	2.493	2.444	2.367	2.317	2.335	2.363	2.382	2.392	2.410	2.433	2.476	2.507	(41)	
(42)		Ebn at Noon	812	857	865	870	868	867	868	859	840	795	772	781	(42)	
(43)		Edh at Noon	72	88	108	122	124	121	117	112	102	89	73	65	(43)	
(44)	All-Sky Solar Radiation	RadAvg	1.81	2.78	4.12	5.39	6.50	7.48	7.47	6.42	4.88	3.10	1.99	1.57	(44)	
(45)		RadStd	0.18	0.27	0.30	0.35	0.45	0.29	0.27	0.20	0.26	0.26	0.22	0.15	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (31 neighbors)	+0.30	N/A	N/A	+0.67	N/A	N/A	N/A	N/A	N/A	+50

Nomenclature: See separate page